MIL-T-23907D
18 August 1987
SUPERSEDING
MIL-T-23907C
1 August 1980

MILITARY SPECIFICATION

TOWING AND DISCHARGE ASSEMBLIES, AMPHIBIOUS FUEL LINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an assembly of cable, pipe, fittings, and accessories, as required to make up a pipeline towing arrangement, a pipeline retrieving arrangement, a flexible pipeline to tanker hose connection, and an inshore casing to pipeline to tanker hose connection and an inshore casing to pipe adapter connection for 6-inch fuel lines.

1.2 Part Number. The specification part number for the towing and discharge assemblies covered by this specification shall be M23907-01.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-C-450 - Clamp, Wire Rope.
GG-G-76 - Gages, Pressure and Vacuum, Dial Indicating, (For Air, Steam, Oil, Water, Ammonia and Cloro-Fluorohydrocarbon Gases, and Compress Gases).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3835

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RR-S-550 - Sockets, Wire Rope.
RR-W-410 - Wire Rope and Strand.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-V-12003 - Valves, Plug: Cast-Iron or Steel, Manually Operated.
MIL-T-22085 - Tape, Pressure-Sensitive Adhesive, Preservation and Sealing.

STANDARDS

FEDERAL

FED-STD-H28/2 - Screw-Thread Standards for Federal Services, Section 2, Unified Inch Screw Threads UN and UNR Thread Forms.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

2.1.2 Government drawings. The following Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING

Naval Facilities Engineering Command (NAVFAC)
(Code Ident 80091)

6028008 - Pipeline F/ 6 Inch Amphibious Assault Submarine Fuel Line.

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

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AMERICAN PETROLEUM INSTITUTE (API)

5A - Casing, Tubing, and Drill Pipe, Specification for.
5B - Threading, Gaging and Thread Inspection of Casing, Tubing, and Line Pipe Threads, Specification for.

(Application for copies should be addressed to the American Petroleum Institute, 1220 L. Street, NW, Washington, DC 20005.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A242 - High-Strength Low-Alloy Structural Steel.
D3951 - Packaging, Commercial.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.2 First article. When specified in the contract or purchase order, a sample shall be furnished for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.3 Design and construction. Towing and discharge assemblies and parts shall be constructed in accordance with NAVFAC Drawing 6028008 and as specified herein. Zinc, zinc-coated steel, copper and copper bearing alloys shall not be used in parts which contact aviation fuels.

3.3.1 Assemblies, adapters, flanges, and eyes. Material used for assemblies, adapters, flanges, and eyes shall be constructed of steel conforming to API 5A and ASTM A242 class 150 as applicable. Screw threads for pipeline coupling connections shall be buttress threads conforming to API 5B. Screw threads for hose adapter couplings shall conform to FED-STD-H28/2, form as indicated on NAVFAC Drawing 6028008.

3.3.2 Messenger cable assemblies. Wire rope used in the messenger cable assemblies shall conform to RR-W-410, type I, class 2, 6 by 19 extra improved plow steel, galvanized, size 7/8 inch, 79,600 pound-force nominal breaking strength. The rope sockets shall conform to RR-S-550, type A, finish 2. Wire rope clips shall conform to FF-C-450, type I, class 2.

3.3.3 Plug valve. The plug valve shall conform to MIL-V-12003, type II, style B, class 125, size 6 inch.

3.3.4 Pressure gage. The pressure gage shall conform to GG-G-76, type I, class 1, style A, range 0 to 200 pound-force per square inch (psi). The gage shall have an integral shutoff cock or valve or be provided with a 1/4-inch nominal size valve, close nipple, and coupling.

3.4 Performance. When the assemblies are arranged as shown in NAVFAC Drawing 6028008 and attached to a pipeline coupling used with 6-5/8 inch outside diameter casing at 20 pounds per foot, the length between the center lines of the messenger cable assembly eyes shall be 202 feet 2 inches. With the bridle of the towing assembly held in a fixed position, the arrangement shall be capable of withstanding a sustained pull at the coupling of 50,000 pound-force at an angle of 15 degrees in any direction without damage or leaking. The arrangement shall be capable of withstanding a sustained pull between the towing assembly and the intermediate adapter of not less than 90,000 pound-force without permanent deformation of any parts.

3.5 Identification marking. Identification shall be permanently and legibly marked directly on the towing and discharge assemblies or on a corrosion-resisting metal plate securely attached to the towing and discharge assemblies at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.6 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill

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scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.7 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.8 Workmanship.

3.8.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.8.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.8.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one towing and discharge assembly when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.1 and 4.4.3, and the packaging inspection of 4.5.

4.3 Examination. Each towing and discharge assembly shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.4 Tests.

4.4.1 Hydrostatic pressure test. The towing and discharge assemblies shall be arranged in a towing arrangement as shown on NAVFAC Drawing 6028008 and with the connecting assembly attached to the towing assembly. A measured hydrostatic pressure of 200 psi shall be applied and maintained for not less than 5 minutes. During the test, the plug valve shall be opened and closed not less than two times. Before reducing pressure, the plug valve

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shall be closed and the cap slowly removed to determine leakage to the valve. Leakage at connections, inaccuracy of the gage or inability of the valve to maintain pressure shall constitute failure of this test.

4.4.2 Performance test. With the bridle of the towing assembly held in a fixed position, the arrangement shall be capable of withstanding a sustained pull at the coupling of 50,000 pound-force without damage or leaking. The arrangement shall be capable of withstanding a sustained pull between the towing assembly and the intermediate adapter of not less than 90,000 pound-force without permanent deformation of any parts. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.2.1 Lugs. With towing assembly and the intermediate adapter securely mounted, the lugs for the messenger cable shall be tested by the application of a 50,000 pound-force pull on each lug with properly installed wire rope. Tension shall be applied parallel to the section centerline and at 15 degrees from the centerline in both directions. Tension shall be sustained at measured load for not less than 1 minute.

4.4.2.2 Towing assembly. With the towing assembly securely mounted in a manner not to permit rotating, a pull of 90,000 pound-force shall be applied to the bridle along the centerline of the assembly for a period of not less than 5 minutes. Pull of 50,000 pound-force shall be applied to the bridle along the centerline of the assembly and 15 degrees from the centerline to both sides, with the bridle in the inline position and perpendicular position. Each pull shall be sustained for not less than 5 minutes.

4.4.2.3 Messenger cables. The towing and discharge assemblies shall be assembled in the towing arrangement as indicated on NAVFAC Drawing 6028008 except the retrieving eye shall be assembled in place of the inshore adapter. The tension of 1,000 pound-force shall be applied and the distance between the eyes of the messenger cable shall be measured to determine conformance to 3.5. The tension shall be increased to 90,000 pound-force and sustained for not less than 5 minutes. The tension shall then be reduced to 1,000 pound-force and the distance between the messenger cable eyes remeasured. Any permanent stretch of the messenger cable shall constitute failure of this test.

4.4.3 Screw threads. All screw threads shall be tested with "go/no-go" gages for conformance to the tolerances specified in FED-STD-H28/2 and API 5B, as applicable.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

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5.1.1 Level A. Preservatives shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.1 Unpainted surfaces. Unpainted exterior surfaces, including threaded surfaces, shall be coated with type P-1 preservative.

5.1.1.2 Valves, adapters, piping, and assemblies. Interior surfaces of ferrous metal valves, adapters, piping, and assemblies shall be coated with type P-10 preservative. Exposed threaded surfaces not protected by caps shall be covered with tape conforming to MIL-T-22085, type II.

5.1.1.3 Gages. Gages shall be preserved in accordance with MIL-P-116, Method IC.

5.1.2 Level C. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations. The complete set of towing and discharge assemblies shall be preserved in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The towing and discharge assemblies shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes and damaging of the contents.

5.2.2 Level B. The towing and discharge assemblies shall be packed as specified for level A, except that the boxes shall be domestic type.

5.2.3 Level C. The towing and discharge assemblies shall be packed in accordance with ASTM D3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The towing and discharge assemblies are intended for use in the installation and removal by towing of 6-inch bottom laid ship-to-shore bulk fuel delivery pipelines, and surface pipeline connections at tank farms.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Color of finish coat (see 3.6).
c. When first article sample is required for inspection and approval (see 4.2.1).
d. Level of preservation required (see 5.1).
e. Level of packing required (see 5.2).

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6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DOD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DOD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Inshore casing to pipe adapter connection
Messenger cable assemblies
Pipeline retrieving arrangement
Tanker hose connection
Towing and discharge assemblies, amphibious fuel line

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

Review Activities:
Army - AT
DLA - CS

Preparing activity:
Navy - YD

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-23907D	TOWING AND DISCHARGE ASSEMBLIES, AMPHIBIOUS FUEL LINE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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